

Human Nutrition Science

Macronutrients, Micronutrients, Metabolism, and Public Health

1. Introduction to Nutrition Science

Nutrition science is the study of how food and its components affect the health, growth, reproduction, and survival of living organisms. For humans, adequate nutrition is fundamental to physical development, cognitive function, immune competence, and the prevention of chronic disease. Despite remarkable advances in nutritional knowledge over the past century—from the identification of vitamins that cured deficiency diseases to the discovery of the gut microbiome's profound influence on human health—nutrition remains one of the most complex and contested fields in biomedical science. Diet-related chronic diseases, including obesity, type 2 diabetes, cardiovascular disease, and certain cancers, account for a substantial proportion of global mortality and morbidity, costing trillions of dollars in healthcare expenditure annually.

2. Macronutrients

2.1 Carbohydrates

Carbohydrates are the body's primary and preferred energy source, providing 4 kilocalories per gram. They are classified by their chemical structure into simple sugars (monosaccharides like glucose and fructose, and disaccharides like sucrose and lactose) and complex carbohydrates (starches and dietary fiber, consisting of longer chains of glucose units). Dietary guidelines generally recommend that 45–65% of total caloric intake come from carbohydrates, with an emphasis on complex, fiber-rich sources over refined sugars. The glycemic index (GI) and glycemic load (GL) are measures of how rapidly foods raise blood glucose levels; high-GI diets are associated with increased risk of type 2 diabetes and cardiovascular disease.

Dietary fiber—the non-digestible fraction of plant foods—is of particular importance. Soluble fiber (found in oats, legumes, and fruits) dissolves in water to form a viscous gel, which slows nutrient absorption, reduces post-meal blood glucose spikes, and lowers LDL cholesterol. Insoluble fiber (found in whole grains, vegetables, and nuts) adds bulk to stool and promotes regular bowel movements. Fermentation of fiber by gut bacteria produces short-chain fatty acids (SCFAs) like butyrate, propionate, and acetate, which serve as energy substrates for colonocytes (colon cells), modulate immune function, and may protect against colorectal cancer.

2.2 Proteins

Proteins are macromolecules composed of amino acid chains and serve structural, enzymatic, hormonal, immune, and transport functions throughout the body. Of the 20 amino acids used in

human proteins, 9 are essential—meaning they cannot be synthesized by the body and must be obtained from the diet. Complete proteins (found in animal products like meat, poultry, fish, eggs, and dairy, as well as soy) contain all nine essential amino acids in adequate proportions. Most plant proteins are incomplete, though combining complementary plant sources (such as beans and rice) throughout the day ensures adequate essential amino acid intake for vegetarians and vegans.

The Recommended Dietary Allowance (RDA) for protein is 0.8 grams per kilogram of body weight per day for sedentary adults—sufficient to prevent deficiency but not necessarily optimal for all individuals. Athletes, older adults (to counteract age-related muscle loss or sarcopenia), and those recovering from illness or surgery have higher protein requirements. Emerging research suggests that protein distribution across meals matters: consuming 25–40 grams of high-quality protein per meal maximally stimulates muscle protein synthesis, while the remainder of daily intake may be used primarily for energy or oxidized.

2.3 Fats

Dietary fats provide 9 kilocalories per gram—more than twice the energy density of carbohydrates or protein—and serve as structural components of cell membranes, precursors for hormones and signaling molecules, carriers for fat-soluble vitamins (A, D, E, K), and substrates for the synthesis of eicosanoids (prostaglandins, thromboxanes, leukotrienes) that regulate inflammation and immune responses. The health effects of dietary fat have been one of the most debated topics in nutrition science.

Saturated fats, found in animal products and tropical oils (coconut, palm), raise LDL cholesterol and are associated with increased cardiovascular risk in most epidemiological studies, though the relationship is modified by the specific fatty acids and the foods they replace in the diet. Trans fats, created through industrial partial hydrogenation of vegetable oils, have been conclusively linked to cardiovascular disease and have been largely banned from food supplies in many countries. Unsaturated fats—both monounsaturated (found in olive oil, avocados, and nuts) and polyunsaturated (including omega-6 and omega-3 fatty acids)—are generally considered healthful. Omega-3 fatty acids (found in fatty fish, flaxseed, and walnuts) have anti-inflammatory properties and are associated with reduced cardiovascular risk.

3. Micronutrients: Vitamins and Minerals

Micronutrients—vitamins and minerals—are required in relatively small amounts but are essential for a vast array of metabolic functions. Vitamins are organic compounds classified as either fat-soluble (A, D, E, K) or water-soluble (B-complex vitamins and vitamin C). Minerals are inorganic elements required for structural purposes (calcium and phosphorus in bone), enzyme function (zinc, selenium, iron), fluid balance (sodium, potassium, chloride), and many other roles. Globally, micronutrient deficiencies—sometimes called "hidden hunger"—affect over 2 billion people. Iron deficiency anemia affects approximately 1.2 billion people worldwide and is the most common nutritional disorder; it causes fatigue, impaired cognitive function, and increased maternal

and infant mortality.

Vitamin D, synthesized in the skin through UV-B radiation exposure and obtained through diet (fatty fish, fortified foods), plays critical roles in calcium absorption and bone mineralization, immune function, and cell proliferation. Vitamin D deficiency is highly prevalent globally—estimated to affect over 1 billion people—particularly in populations with limited sun exposure, darker skin pigmentation, or older age. Folate (vitamin B9) is essential for DNA synthesis and cell division; deficiency during early pregnancy significantly increases the risk of neural tube defects, which is why folic acid fortification of grain products and supplementation recommendations for women of childbearing age are important public health measures.

4. The Gut Microbiome

The human gut harbors approximately 38 trillion bacteria, along with archaea, fungi, and viruses, collectively known as the gut microbiome. This complex ecosystem plays fundamental roles in digestion (particularly of dietary fiber), vitamin synthesis (vitamins K and B12), immune system development and regulation, and protection against pathogenic microorganisms. The composition of the gut microbiome is shaped by diet, age, medications (particularly antibiotics), mode of birth (vaginal vs. cesarean), infant feeding (breastfeeding vs. formula), and early life exposures. Diet is among the most powerful modifiers of the gut microbiome in adults.

A diet rich in diverse plant foods—vegetables, fruits, whole grains, legumes, nuts, and seeds—has been consistently associated with greater microbiome diversity, which is generally considered a marker of a healthy gut. Fermented foods (yogurt, kefir, kimchi, sauerkraut) introduce beneficial live microorganisms (probiotics) and are associated with increased microbiome diversity and reduced inflammatory markers. Dysbiosis—an imbalance in the gut microbiome composition—has been linked to conditions including inflammatory bowel disease, obesity, type 2 diabetes, mental health disorders, and autoimmune conditions, suggesting that the gut microbiome may be an important mediator of the relationships between diet and many chronic diseases.

5. Nutrition and Chronic Disease Prevention

Diet is a major modifiable risk factor for the leading causes of global mortality. Cardiovascular disease (CVD), the world's leading cause of death, is strongly influenced by dietary patterns. The Mediterranean diet—characterized by high intake of olive oil, vegetables, legumes, whole grains, fish, and moderate red wine consumption with limited red meat and processed foods—has the strongest evidence base for cardiovascular protection, reducing risk of major cardiovascular events by approximately 30% in high-risk individuals in the PREDIMED randomized controlled trial. Type 2 diabetes risk is strongly modulated by diet and body weight; diets high in ultra-processed foods, refined carbohydrates, and sugar-sweetened beverages are associated with substantially increased risk, while dietary patterns emphasizing fiber-rich plant foods, whole grains, and healthy fats reduce risk.

6. Conclusion

Nutrition science has transformed our understanding of how what we eat shapes our health across the lifespan. The shift from a focus on individual nutrients to overall dietary patterns reflects growing appreciation for the complexity of food matrices and the interactions between dietary components. Translating nutritional science into practical guidance for individuals and populations remains challenging due to the complexity of diet, individual variation in nutrient metabolism (influenced by genetics, gut microbiome, and other factors), and the difficulty of conducting rigorous long-term dietary intervention studies. Nevertheless, a consistent message emerges from the evidence: diets centered on minimally processed plant foods, adequate in protein and essential fats, and limited in ultra-processed products, added sugars, and excessive sodium provide the strongest foundation for long-term health.

Table 5: Key Vitamins — Functions, Sources, and Deficiency Diseases

Vitamin	Key Functions	Major Food Sources	Deficiency Disease
A (Retinol)	Vision, immune function	Liver, dairy, orange vegetables	Night blindness, xerophthalmia
D (Calciferol)	Calcium absorption, bone health	Fatty fish, fortified milk, sun	Rickets (children), osteomalacia
B12 (Cobalamin)	DNA synthesis, nerve function	Meat, fish, eggs, dairy	Megaloblastic anemia, neuropathy
C (Ascorbic acid)	Antioxidant, collagen synthesis	Citrus fruits, peppers, kiwi	Scurvy
Folate (B9)	DNA synthesis, cell division	Leafy greens, legumes, fortified grains	Neural tube defects, anemia
Iron (mineral)	Oxygen transport (hemoglobin)	Red meat, legumes, fortified cereal	Iron deficiency anemia